

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

What does scale mean?



Density



Size



Performance

Scale considerations

- Capacity
- Performance
- Agility
- Cost efficiency
- Automation
- Ease of management
- Reliability
- Predictability

STG320-R

Best practices for operating at scale with Amazon EBS & EBS snapshots

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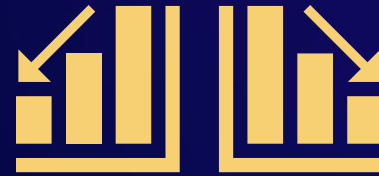
AWS



Whiteboard: Distributed multi-tier application architecture

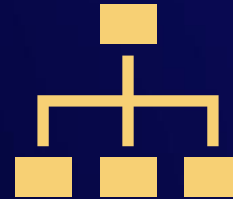
EBS service quotas

What and why



Quota adjustment

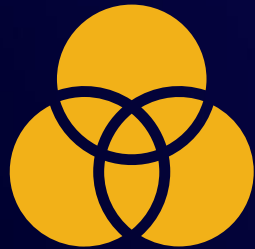
Monitor



Types of quotas

Multiple account benefits

Logical account separation



Security and governance



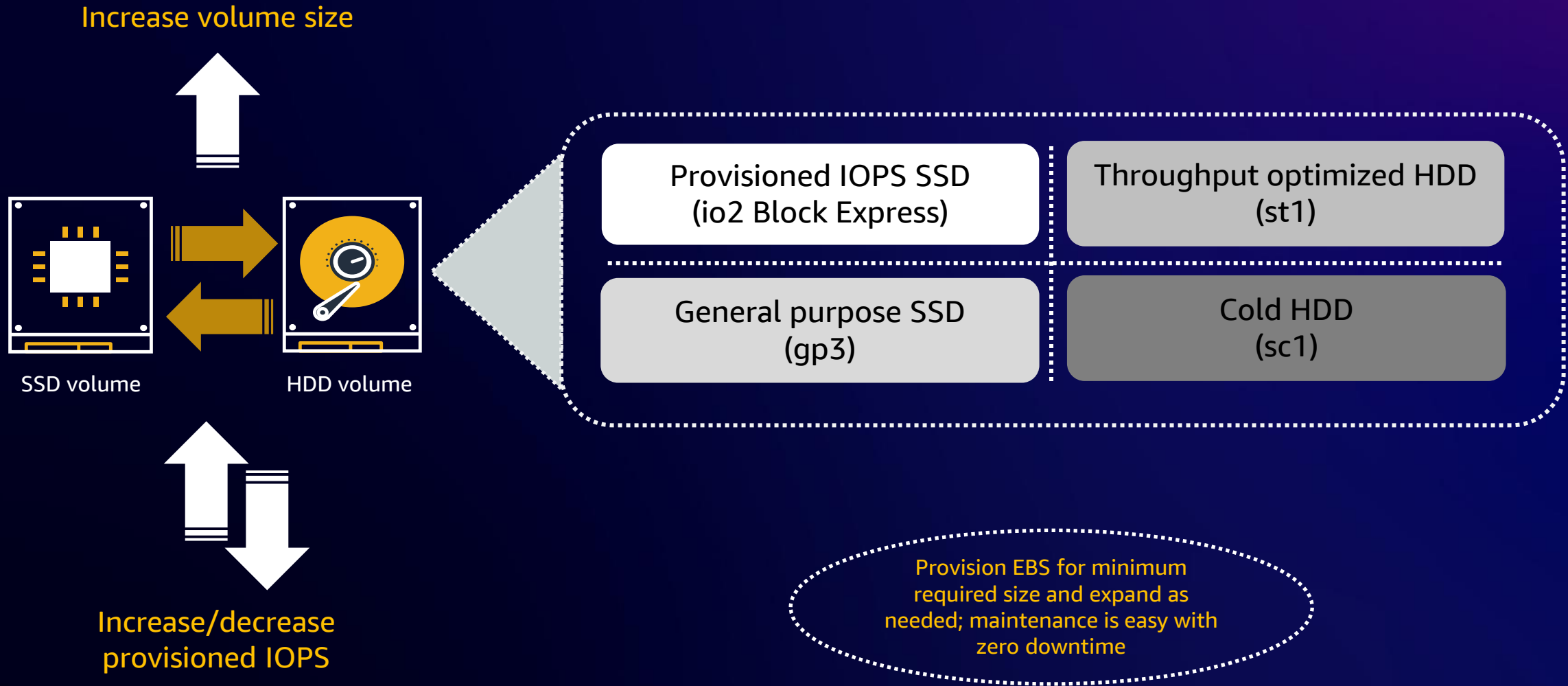
Cost granularity



Scope of impact reduction



Elastic volumes



Preserving performance for gp2 to gp3 EV

Before:

```
awscli: aws ec2 modify-volume $volume-id  
--volume-type gp3 --iops $iops  
--throughput $throughput
```

CloudFormation:

```
"volumeType": gp3  
"Iops": <%= ... %>  
"Throughput": <%= ... %>
```

Terraform:

```
type = "gp3"  
iops = min(max(size*3,16000),3000)  
throughput = min(max(min(max(... ))))
```

Now:

```
awscli: aws ec2 modify-volume $volume-id  
--volume-type gp3
```

CloudFormation: "volumeType": gp3

Terraform: type = "gp3"

 Throughput has been automatically calculated for the gp3 volume to deliver the same performance as the source gp2 volume. You can specify a different value if needed. For more information, please see the [Elastic Volume user guide](#).

Whiteboard: Working with tightly coupled scale systems

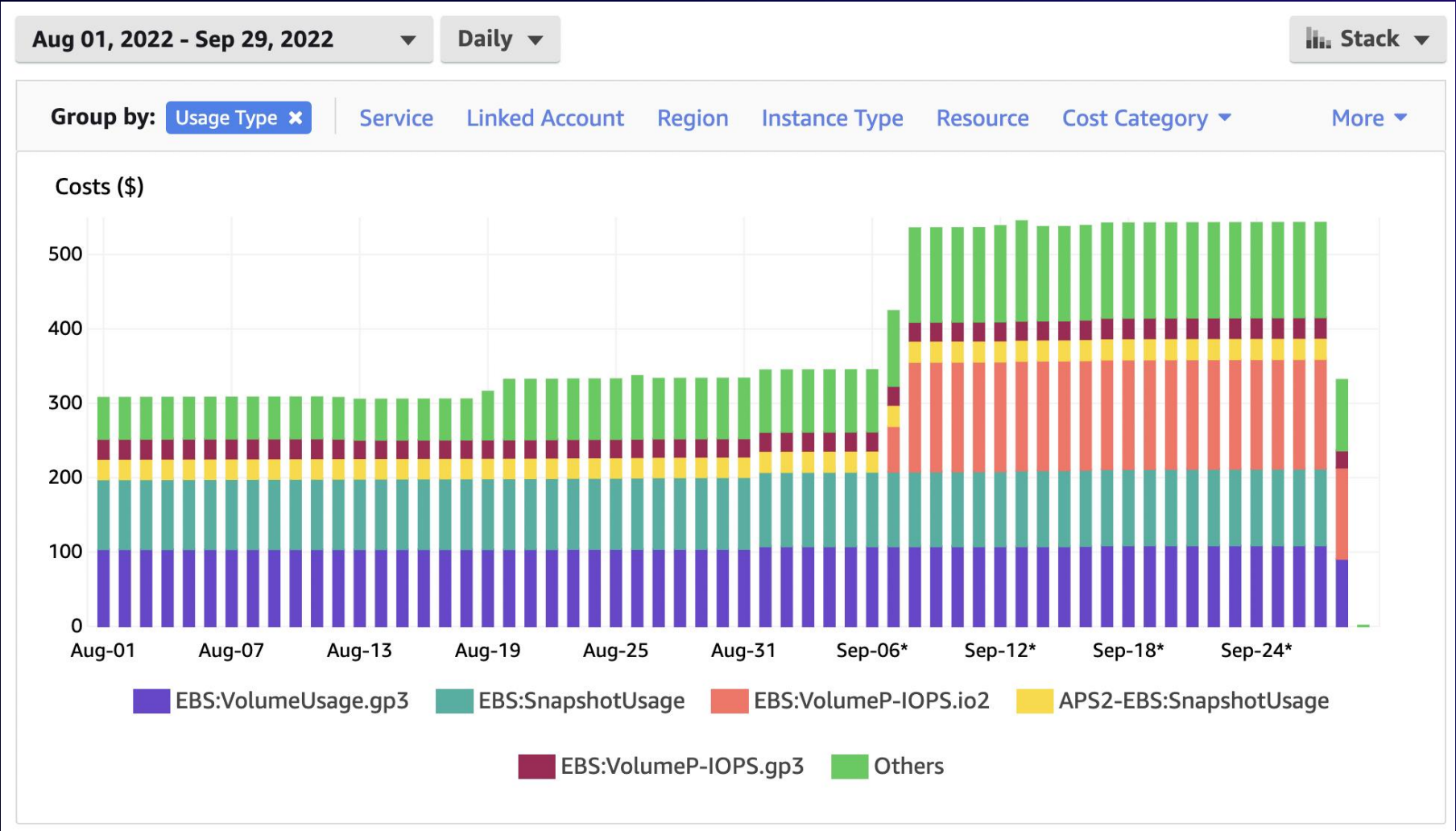
Whiteboard: Migration of 100K gp2 volumes to gp3

Automation

```
for vol in vol_list:
    volume = ec2.volume(vol)
    if volume.volume_type == 'gp2':
        try:
            vol_mod = ec2_client.modify_volume(
                volumeId = vol,
                volumeType = 'gp3'
            )
        except ClientError as e:
            print(vol + ': ' +
e.response['Error']['Message'])
```

<https://github.com/aws-samples/amazon-ebs-migration-utility>

Cost optimization/tagging



Tag

CreatedBy Include all

< CreatedBy

Search for CreatedBy

☐ No Tagkey: CreatedBy

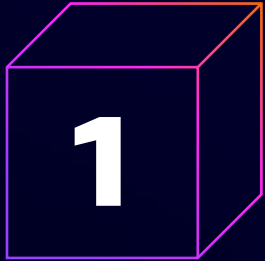
☐ EC2 Image Builder



Whiteboard: Hydrating a subset of a volume

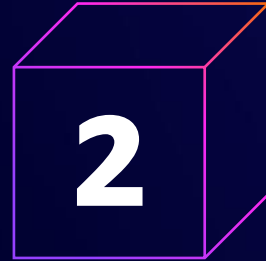
Migration best practices

Key takeaways



Elastic Volumes are non-disruptive

Customers consistently perform large EV migrations with no impact to prod



Be proactive about quotas

Plan ahead and monitor quota utilization



Consider a multi-account strategy

Multiple accounts provide cost granularity, security benefits, and more



Identify your bottlenecks

Proactively monitor and predict system behavior and scaling needs

Interested in learning more about EBS?

FIND OUR NEXT SESSIONS BY SCANNING THE QR CODE



Time	Session name	Type	Location
Tuesday, 11/28, 5:30 – 6:30pm	STG205 – Protect critical data with use using Amazon EBS Snapshots	Breakout	Caesars Forum, Level 1, Academy 407
Wednesday, 11/29, 8:30 – 10:30am	STG349 – Comprehensive guide to deploying business applications with EBS	Workshop	Mandalay Bay, Level 1, North, South Pacific F
Wednesday, 11/29, 11:00 – 11:20am	STG226 – Implement proactive data protection using Amazon EBS Snapshots	Lightning Talk	Venetian, Level 2, Hall B, Expo, Booth#1460, Theater 4
Wednesday, 11/29, 12:00 – 1:00pm	STG332 – Deploying modern applications using Amazon EBS	Chalk Talk	Caesars Forum, Level 1, Academy 411
Wednesday, 11/29, 2:30 – 3:30pm	STG401 – Automate data protection and backup compliance using EBS Snapshots (Repeat)	Builders	Mandalay Bay, Level 2, South, Surf A
Wednesday, 11/29, 5:30 – 6:30pm	STG401 – Automate data protection and backup compliance using EBS Snapshots (Repeat)	Builders	Mandalay Bay, Level 2, South, Surf D
Thursday, 11/30, 11:30am – 1:30pm	STG305 – Build protection and cost-optimize using Amazon EBS	Workshop	MGM Grand, Level 3, Premier 316
Friday, 12/1, 8:30 – 9:30am	STG320 – Best practices for operating at scale with Amazon EBS and Snapshots (Repeat)	Chalk Talk	Caesars, Level 1, Summit 220



Thank you!

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Please complete the session
survey in the mobile app

